MAE 6291
Internet of Things for Engineers

Prof. Kartik Bulusu, MAE Dept.

Week 2 [01/28/2026]	• What is edge computing • What is an edge device • Introduction to the Linux Terminal • Introduction to MQTT • Guest lecture	• Some more programming constructs • RPi skeleton code • Some basic Python programming constructs • Raspberry Pi programming [Blinking LEDs using MQTT approach - Demo] • The Linux Terminal

git clone https://github.com/gwu-mae6291-iot/spring2026_codes.git

GW School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

Spring 2026

Photo: Kartik Bulusu

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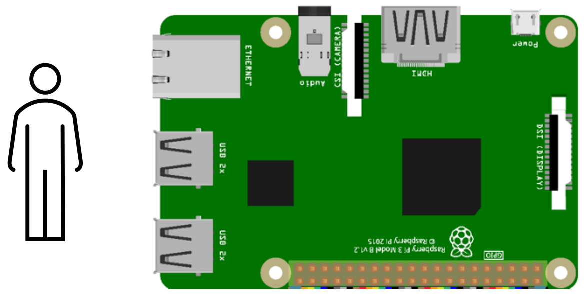
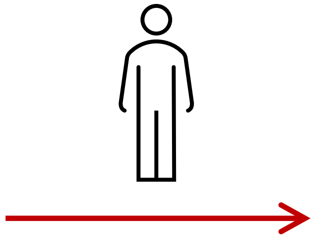
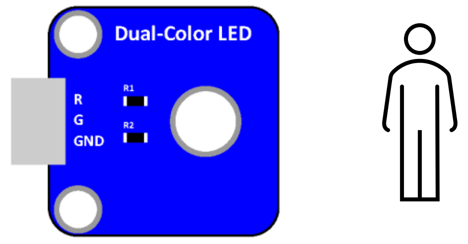
Introduction to MQTT

(Message Queuing Telemetry Transport)

Scenario





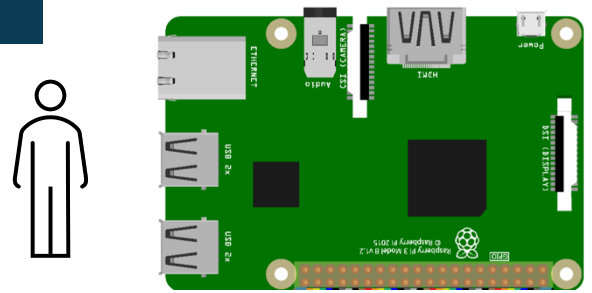


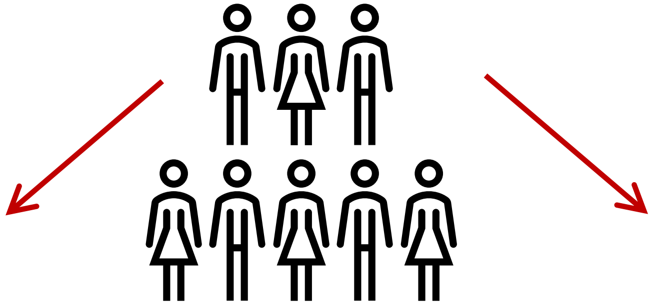
I write some code on a computer and run it

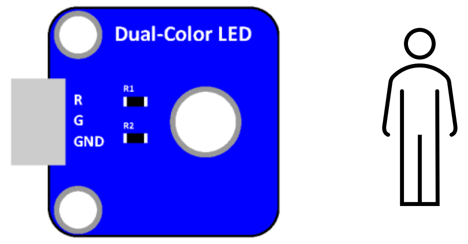
Communicate with a physical entity to do something

Observe response from that entity

What-if-scenario



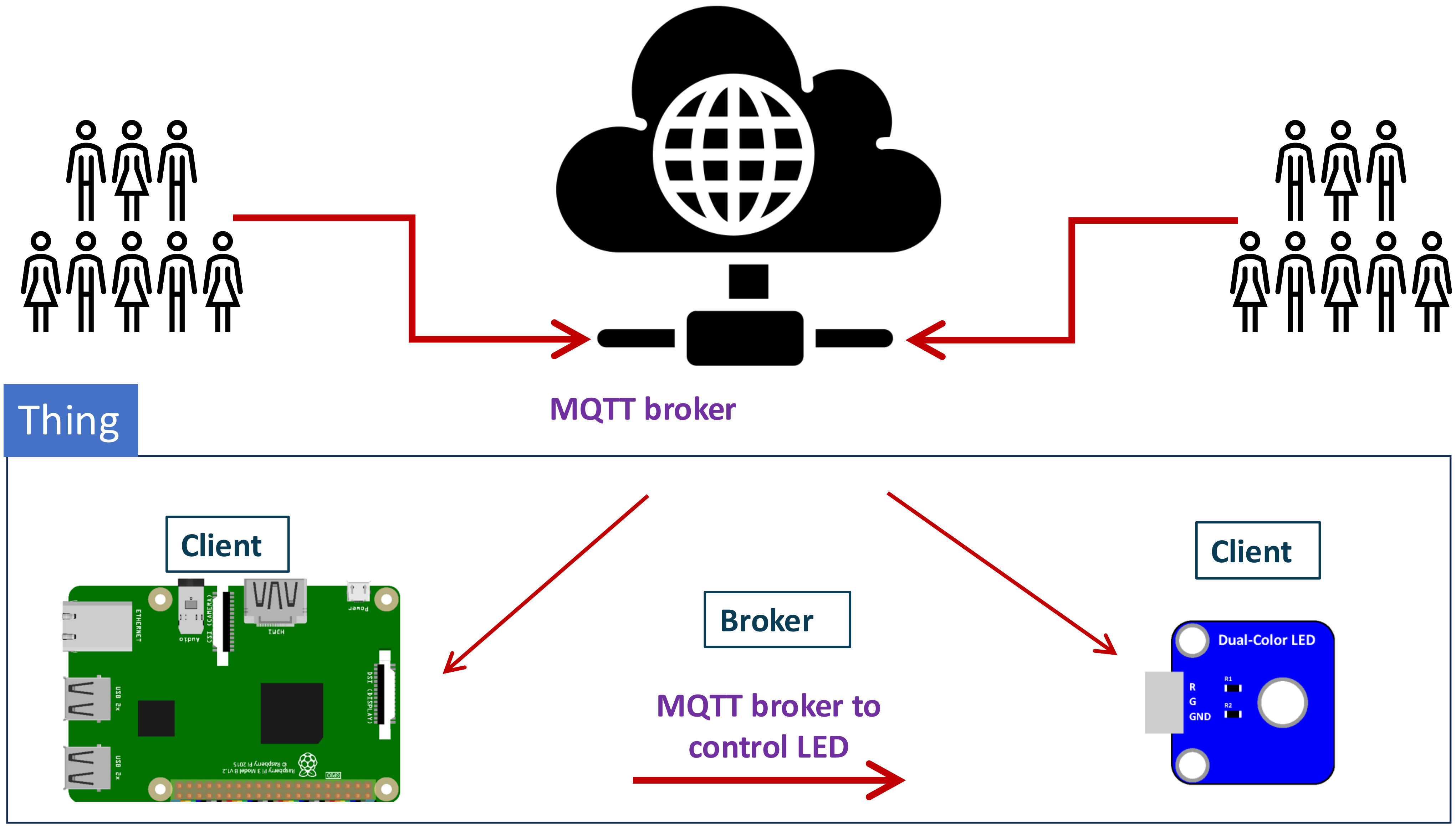




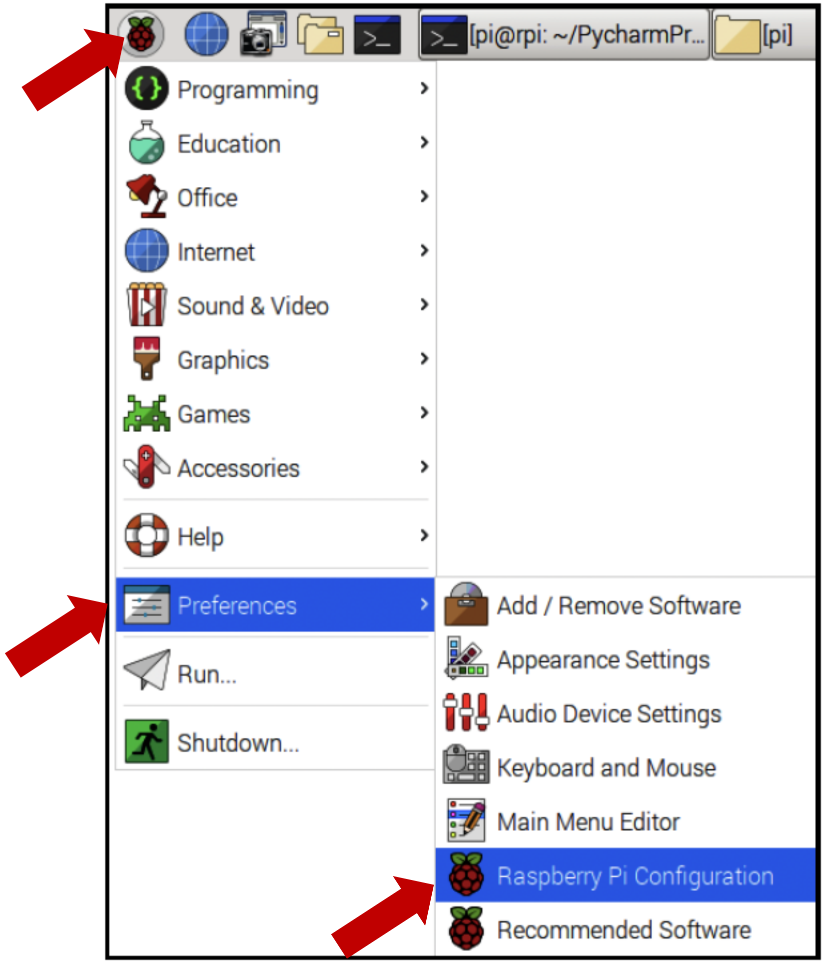
I write some code on a computer

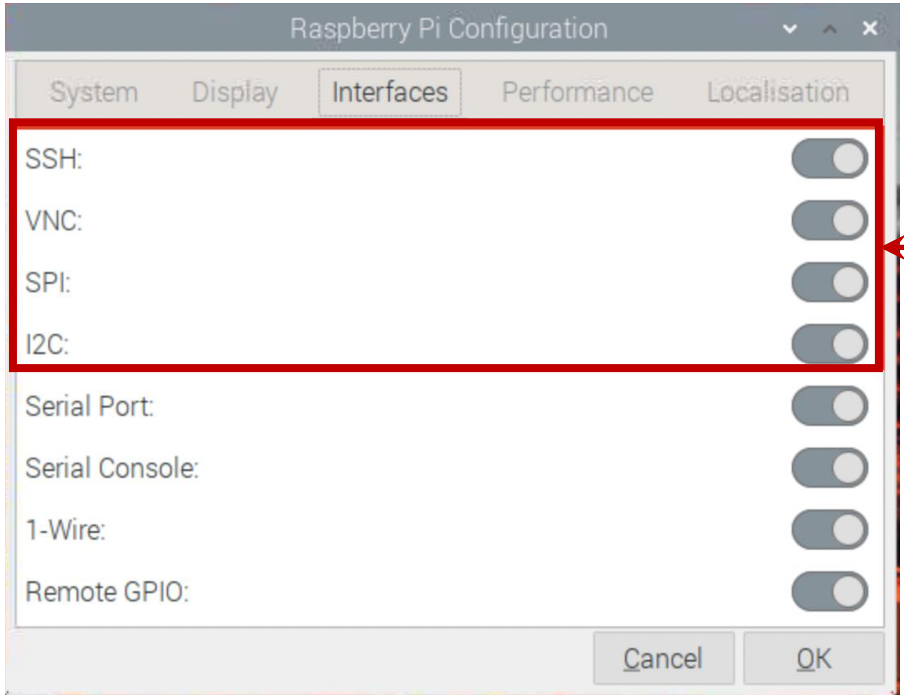
I realize people may not really care about it.
But want to use it anyway.

And curiously observe the response from that entity


MQTT broker
Thing
Client
Broker
Client
Dual-Color LED
MQTT broker to control LED

Couple of checks and balances

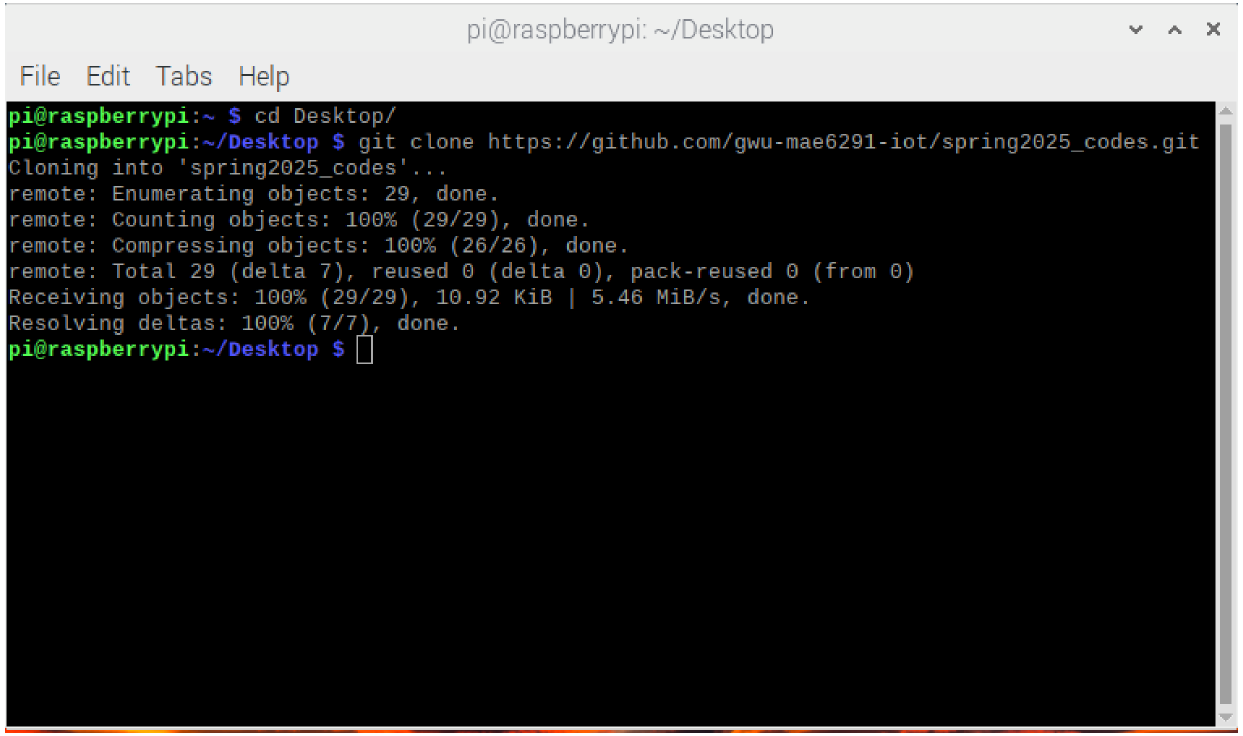


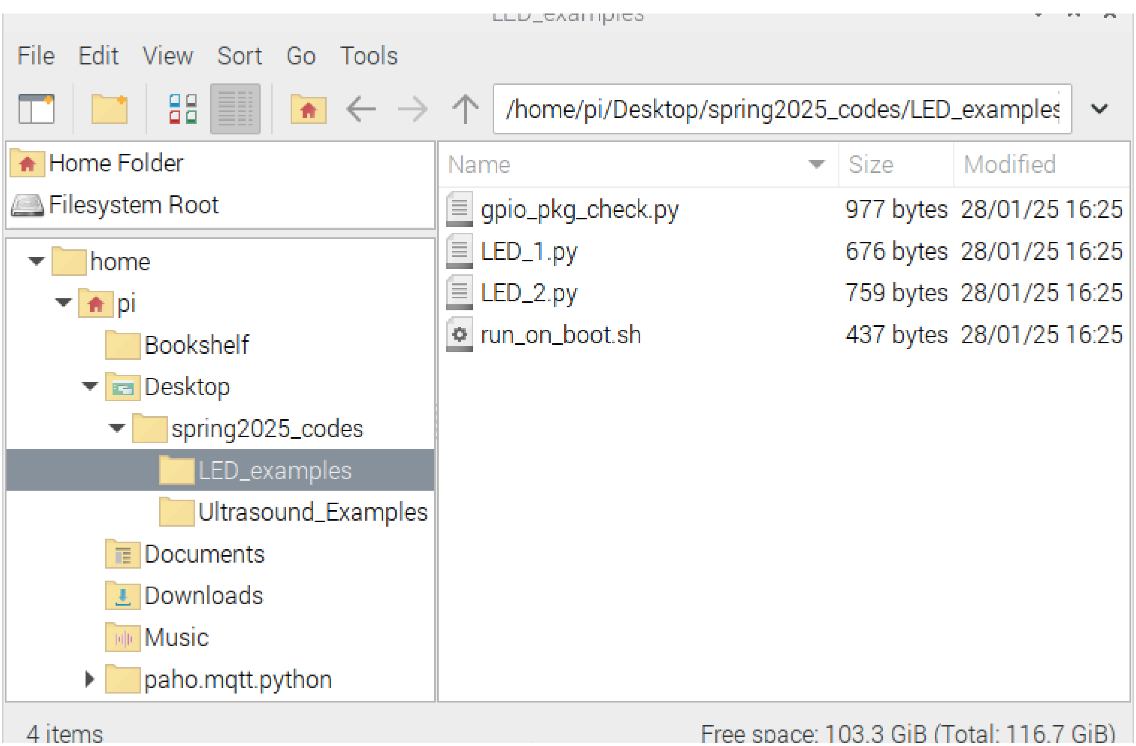


Make sure these are enabled

Downloading folders from course git-repository

$ git clone https://github.com/gwu-mae6291-iot/spring2025_codes.git





“things” with compute

Sources:
Edge computing acts on data at the source: https://www.ibm.com/cloud/what-is-edge-computing
What is Edge computing? https://www.youtube.com/watch?v=3hScMLH7B4o&t=6s

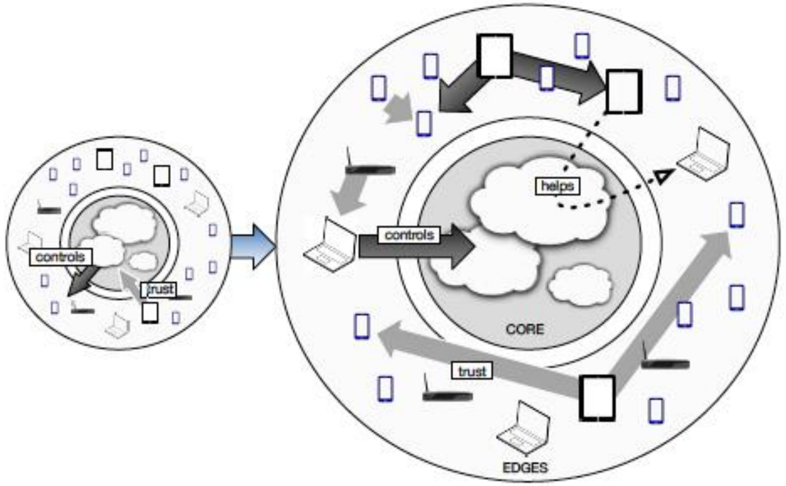
"things" with compute == edge compute

Edge computing is a distributed computing framework that brings enterprise applications closer to data sources such as IoT devices or local edge servers.

This **proximity to data** at its source can deliver strong business benefits, including faster insights, improved response times and better bandwidth availability.

Sources:
Garcia Lopez, Pedro; *et al.*, (30 September 2015). "Edge-centric Computing". *ACM SIGCOMM Computer Communication Review*. **45** (5): 37–42. doi:10.1145/2831347.2831354.
M. Rabinovich, Z. Xiao, and A. Aggarwal. Computing on the edge: A platform for replicating internet applications. In Web content caching and distribution, pages 57{77. Springer, 2004.
Difference Between Edge Computing and Fog Computing, https://www.geeksforgeeks.org/difference-between-edge-computing-and-fog-computing/
Kazaa, https://en.wikipedia.org/wiki/Kazaa#/media/File:Kazaa_(logo).png
Napster, https://en.wikipedia.org/wiki/Napster#/media/File:Napster.png

"Edge Computing" was coined around 2002

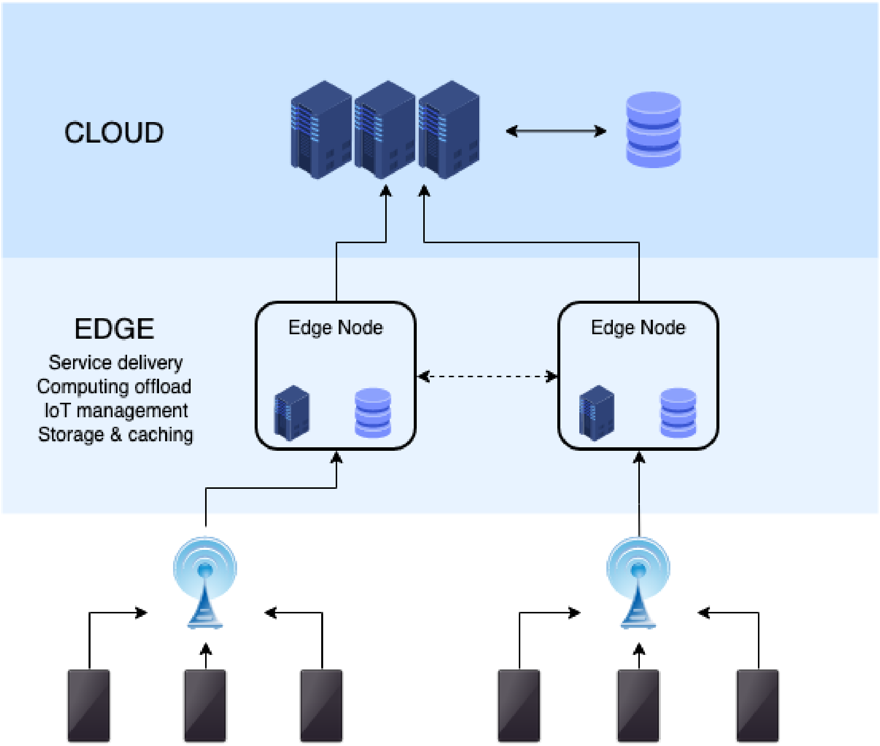


Content Delivery Networks:
It was mainly associated with the deployment of applications over CDNs, when some large companies announced deals to distribute software through CDN edge servers.

P2P computing:
This is another field closely related to edge computing, it is also its main precursor. The term P2P was first introduced around 2000 with the appearance of popular file-sharing systems such as Napster and Kazaa.

Fog computing:
Fog Computing is a recent research field that has substantial overlap with Edge-centric Computing. Proximity to end-users, dense geographical distribution, and support for mobility are the main distinguishing characteristics of Fog Computing.

Edge computing is a form of distributed computing



Classical Paradigm

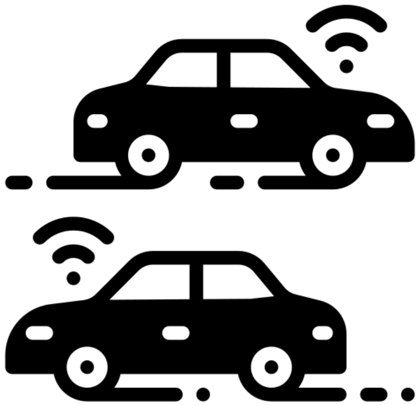
- **Distributed computing**
  - Covers a broad range of technologies
  - Earliest success stories could be considered
    - local area networks and
    - the first internet, ARPANET (1960s).

New Paradigm

- **Decentralized, distributed computing**
  - **Proximity to data:** Moving the computer workload closer to the data source
    - reduces latency
    - bandwidth and
    - overhead for the centralized data center

Icon sources:
Autonomous Car by Tippawan Sookruay from https://thenounproject.com/browse/icons/term/autonomous-car/
clean energy by Juicy Fish from https://thenounproject.com/browse/icons/term/clean-energy
Health Care by mynamepong from https://thenounproject.com/browse/icons/term/health-care/
Smart City by dDara from https://thenounproject.com/browse/icons/term/smart-city/

Four (near future) Edge computing examples

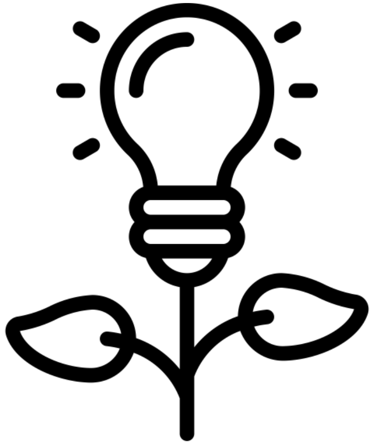


Autononous Cars

- Chevrolet collected 4,220 terabytes of data from customer's cars.

McKinsey forecasts that this could grow into a $450 to 750 billion market by 2030.

Source: https://www.autoblog.com/2017/02/21/race-for-autonomous-cars-is-over-mcelroy-autoline-opinion/

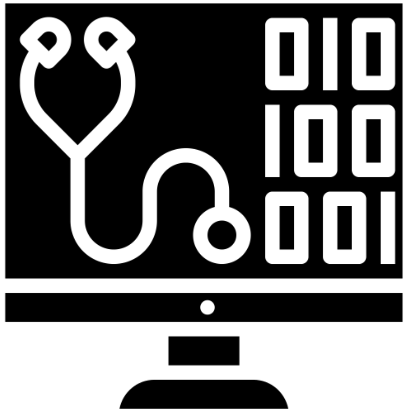


Clean energy technology

- Data centers use an estimated 200 terawatt hours (TWh) of electricity annually
- ~ 50% of all electricity currently used for all global transport.

Edge computing can significantly reduce the amount of time and power, data centers need to use to process data.

Source: https://www.forbes.com/sites/forbestechcouncil/2022/03/18/how-machine-learning-and-edge-computing-powers/ustainability/?sh=483ebd025fab

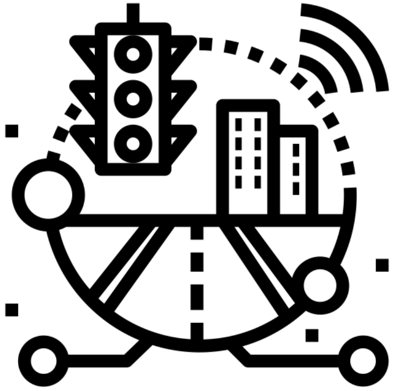


Transforming health care at the edge

- 10 to 15 connected device per US hospital bed
- 3 million data points generated by the average clinical trial
- 30% of all global stored data is from health care

75% of data will be generated at he edge by 2025

Source: https://www.technologyreview.com/2021/06/10/1026038/transforming-health-care-at-the-edge/



Edge AI: Tackling Traffic Management

- A pilot system deployed at Pittsburgh, Pennsylvania, has reportedly
  - reduced travel time by 26%
  - idling time by 41%, and
  - emissions by 21 %.

The INRIX Global Traffic Scorecard:
World's 20 most congested cities lost between 164 and 210 hours in congestion per capita through 2018.

Source: https://www.iotforall.com/busting-traffic-woes-with-5g-and-edge-ai

Sources:
Edge computing, https://en.wikipedia.org/wiki/Edge_computing#/media/File:Edge_computing_infrastructure.png
What is Edge Computing: The Network Edge Explained, https://www.cloudwards.net/what-is-edge-computing/
What is IoT Edge computing? https://www.redhat.com/en/topics/edge-computing/iot-edge-computing-need-to-work-together
What is edge computing?, https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-edge-computing/
Secure Edge Computing in IoT Systems: Review and Case Studies, https://par.nsf.gov/servlets/purl/10097337

Benefits of Edge computing

- **Reduced latency of communication** between IoT devices and the central IT networks.
- **Faster response times** and increased operational **efficiency.**
- **Improved network bandwidth.**
- **Continued systems operation offline** when a network connection is lost.
- **Local data processing**, aggregation, and rapid decision making via analytics algorithms and machine learning.

There are concerns!

- **User Privacy** between IoT devices and the central IT networks.
- **Optimization metrics:** There are several layers with various computation abilities in edge computing for choosing an optimal workload allocation.
- **Task-offloading:** Utilizing edge nodes for computation offloading is a concern due to the problem of adequately segmenting computational tasks.
- **Public accessibility of edge nodes:** When an edge device (e.g., a base station, switch, and router) is intended to be used for public access.

Sources:
Calculator by Markus from https://thenounproject.com/browse/icons/term/calculator/
industrial transformation by dDara from https://thenounproject.com/browse/icons/term/industrial-transformation
Definition: Edge Device, https://www.techtarget.com/searchnetworking/definition/edge-device

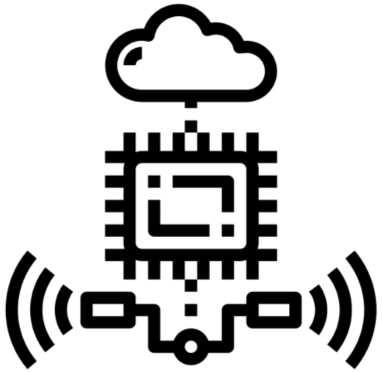
What's an "edge" device?

Paradigm #1

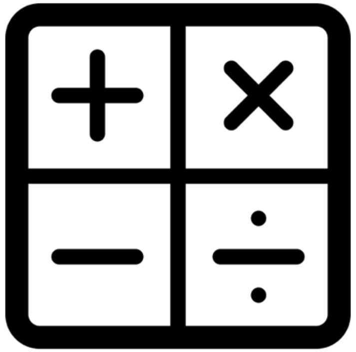
An **edge device** is any piece of hardware that controls data flow at the boundary between two networks.

- Essentially serve as network entry -- or exit -- points.
- Common functions of edge devices are the transmission, routing, processing, monitoring, filtering, translation and storage of data passing between networks.

Paradigm #2



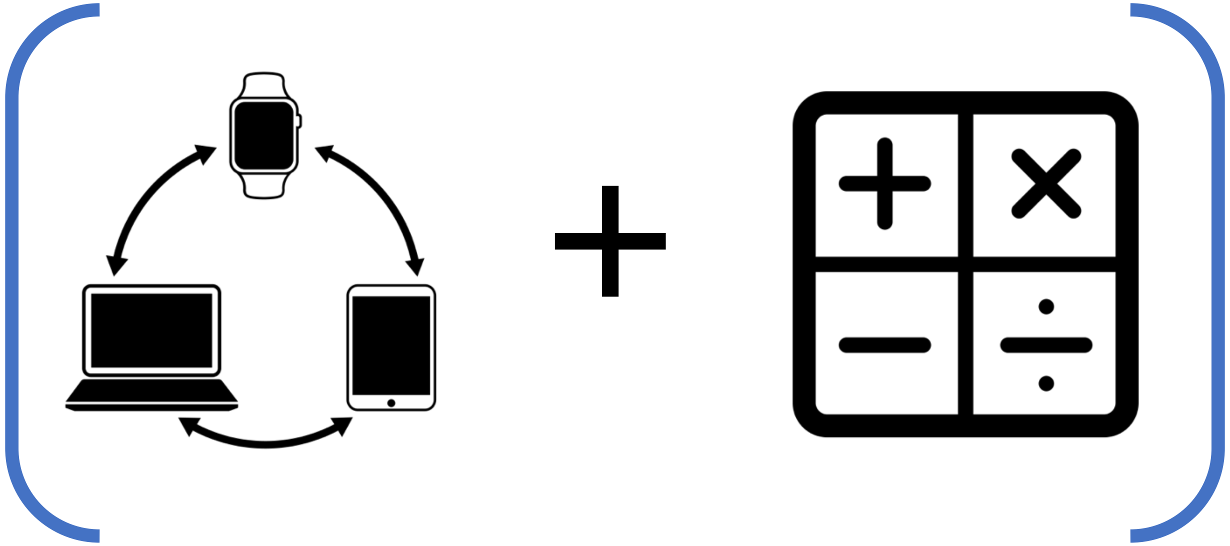
+

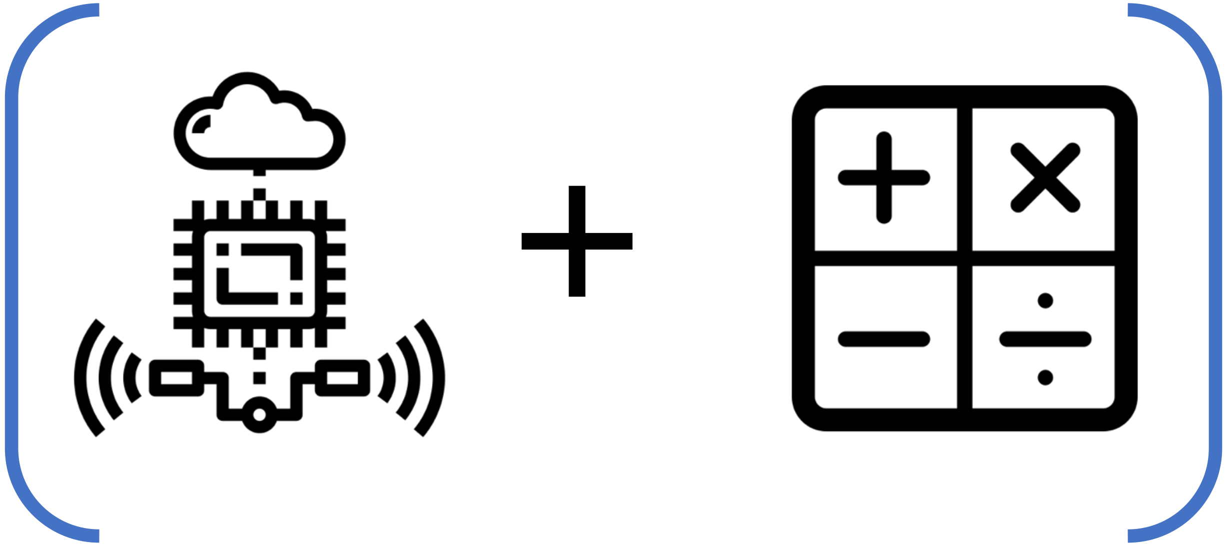


Paradigm #3

Sources:
internet of things by Davo Sime from https://thenounproject.com/browse/icons/term/internet-of-things/
Calculator by Markus from https://thenounproject.com/browse/icons/term/calculator/
industrial transformation by dDara from https://thenounproject.com/browse/icons/term/industrial-transformation
What is IoT Edge computing?, https://www.redhat.com/en/topics/edge-computing/iot-edge-computing-need-to-work-together

Overview of Computing in IoT



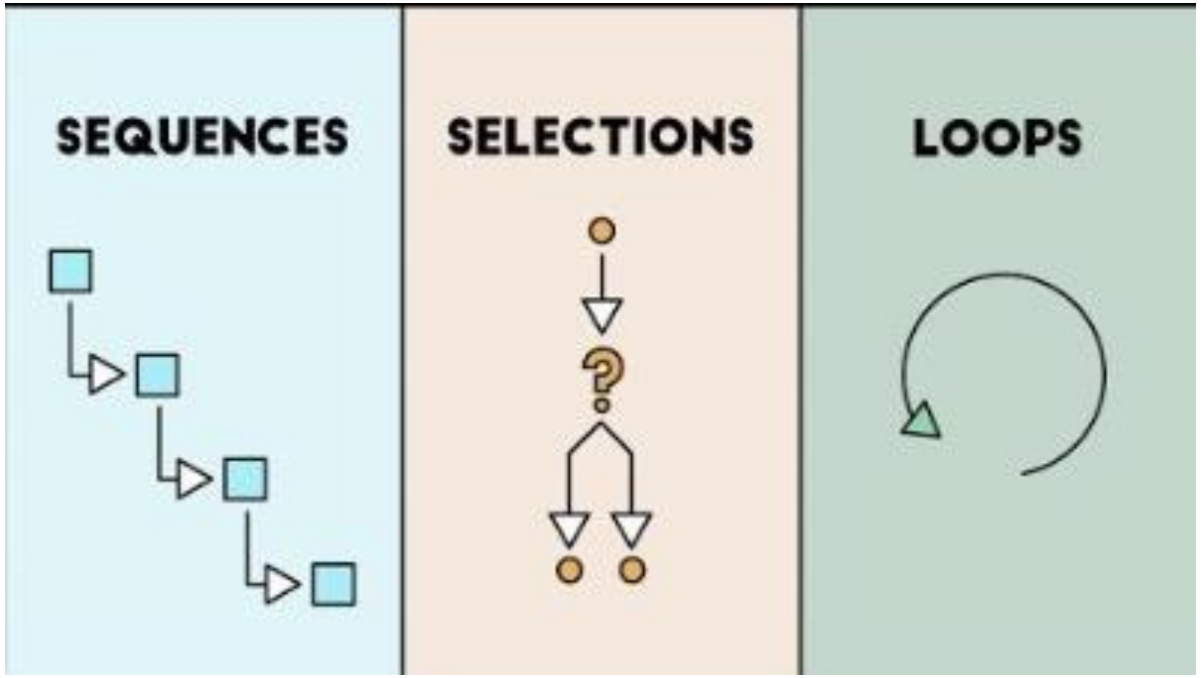


The Internet of Things (IoT) is made up of smart devices connected to a network—sending and receiving large amounts of data to and from other devices—which produces a large amount of data to be processed and analyzed.

Edge computing, a strategy for computing on location where data is collected or used, allows IoT data to be gathered and processed at the edge, rather than sending the data back to a datacenter or cloud.

Let's get to know some programming paradigms

Sources:
Computer Science Basics: Sequences, Selections, and Loops: https://youtu.be/eSYeHlwDCNA



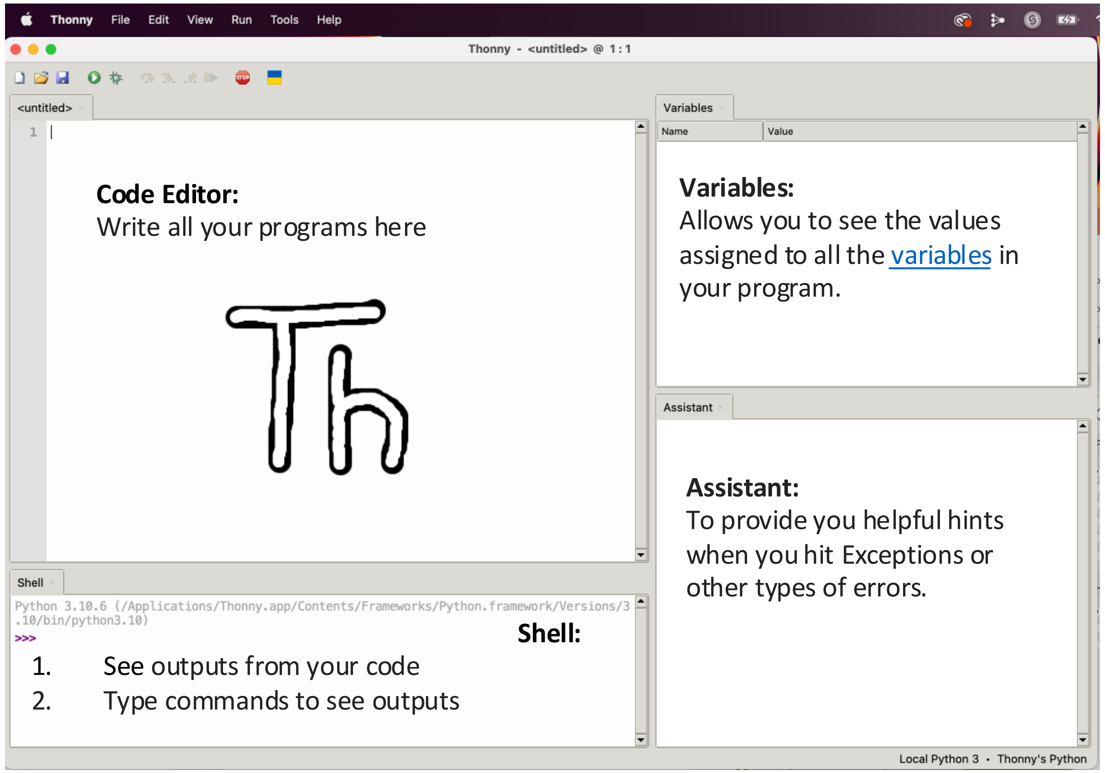
Sequences

Selections

Loops

Thonny integrated development environment (IDE)

Sources:
Thonny (IDE): https://en.wikipedia.org/wiki/Thonny
Thonny: The Beginner-Friendly Python Editor: https://realpython.com/python-thonny/



Allows you to run the code, i.e., "Do what I told you do do!".

Allows you to debug your code.
A bug is another name for a problem.

Allows you to open a file that already exists on your computer

Save your code. Press this early and often.

Create a new file

The arrow icons allow you to run your programs step by step.
These icons are used after you press the bug icon

The resume icon allows you to return to play mode from debug mode.

The stop icon allows you to stop running your code

Built-in function range()

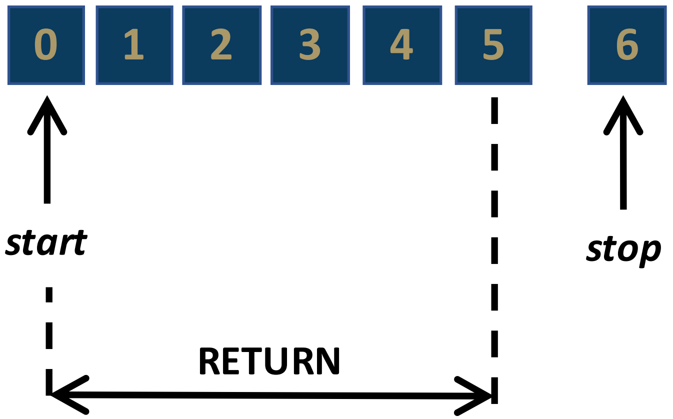
Python's range()function

- returns a sequence of numbers
- works only with integers

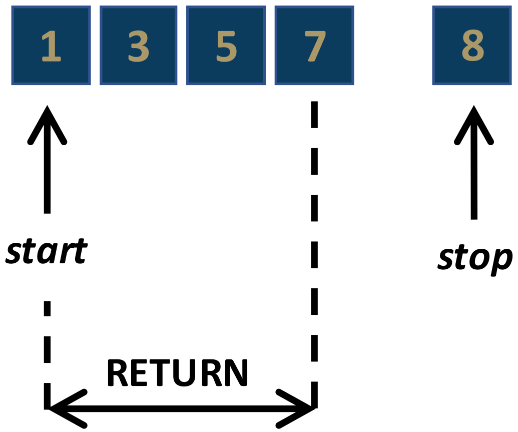
start:	at the value (default = 0)
step:	up or down at the increment value (default = 1)
stop:	at the value but not including it

range(stop) **range(start, stop)** **range(*start*, *stop*, *step*)**

>>> range(6)



>>> range(1,8,2)



range()

- can be utilized in (for) loops
- to specify a range to iterate or do repetitions

Demos

Syntax and Skeleton of a user-defined function

```
def name(parameters):

        statement
        statement
        . . .

        return value
```

Functions are blocks of resuable pieces of code

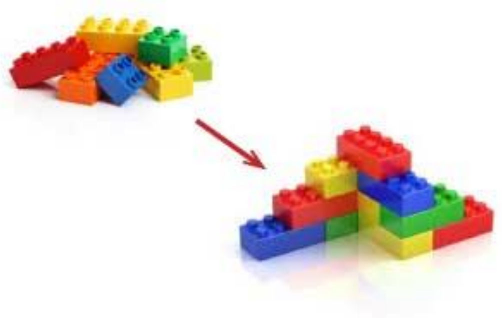


Image source:
https://www.thescoopformommies.com/teaching-colors-to-preschoolers/lego-blocks-clip-art-uiudyc-clipart/
https://www.123rf.com/clipart-vector/lego.html

Function name: Identifier by which it is called in the program

(Optional) **Arguments: values passed to the function**

Function Declaration: Starts with "def" that is not indented

```
def func_name(parameters):

    statement
    statement
    . . .

    return value
```

Colon; Don't miss it!

Indentation: Tab or 4 spaces for each statement

Body: Statements executed each time a function is called

(Optional) **return value: Can end function call and send data back to the main program**

Function definition

```
func_name()
```

Function call

Sources:
Codecademy: https://youtu.be/wxds6MAtUQ0

What is a "loop" in computer programming ?

Repeating routines

From function to repeating functions

Stopping condition

Skeleton of the for-loop

```
for i in range(start, stop, step):

        <expression>
        <expression>
        . . .
```

Range of values to iterate

Default = 0

special word
“in”

stop_value-1

Colon;
Don’t miss it!

```
for i in range(start, stop, step):
        <expression>
        <expression>
        . . .
```

Indentation: Tab
or 4 spaces for
each statement

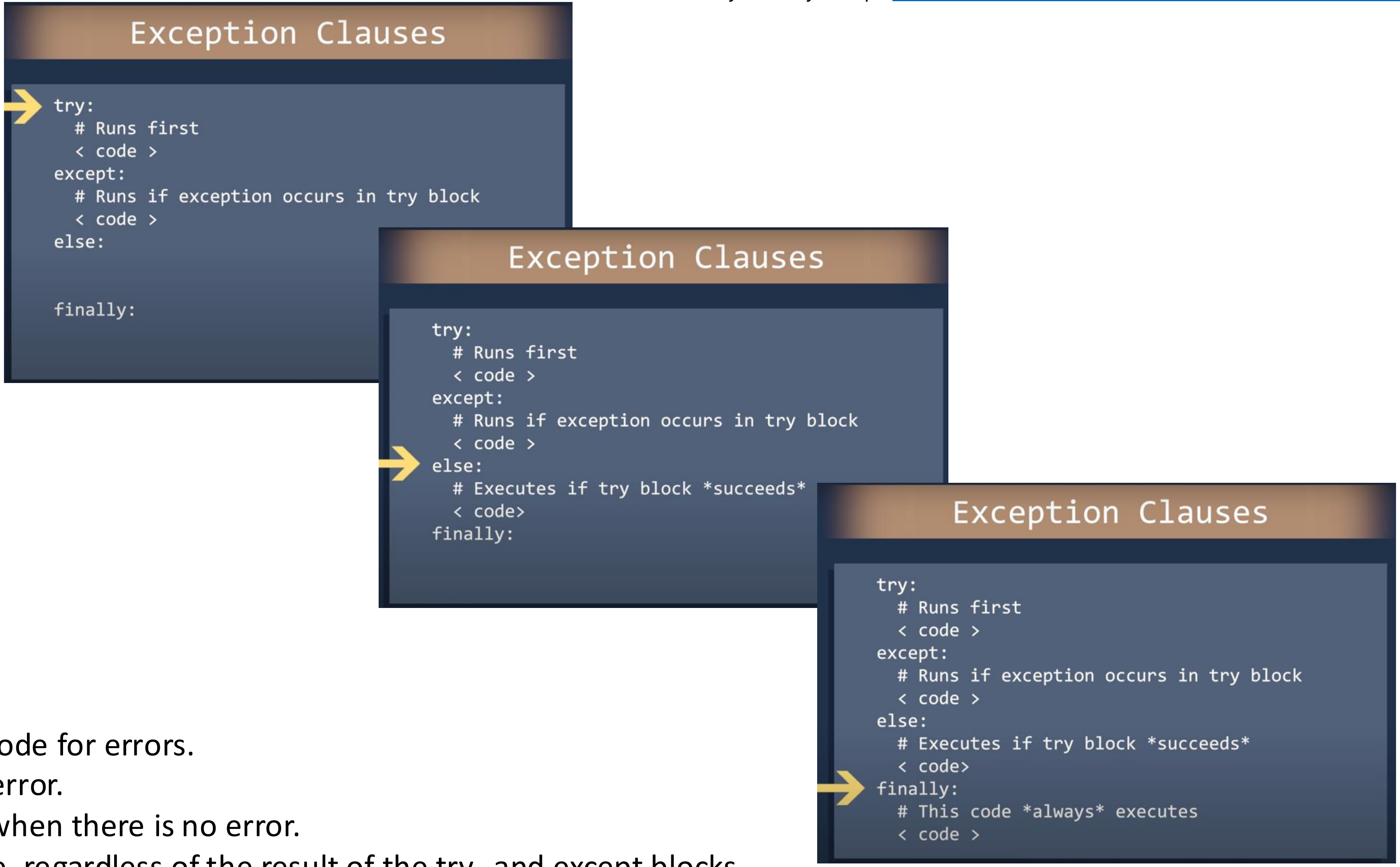
Default = 1

Code block: Expressions executed
within each
iteration

Sources:
Exceptions in Python || Python Tutorial || Learn Python Programming: https://youtu.be/nlCKrKGHSSk
Python Try Except: https://www.w3schools.com/python/python_try_except.asp

Skeleton of
try:
except:
else:
finally:

try:
statement
function
except KeyboardInterrupt:
statement
function
else:
statement
function
finally:
statement
function



The try block lets you test a block of code for errors.
The except block lets you handle the error.
The else block lets you execute code when there is no error.
The finally block lets you execute code, regardless of the result of the try- and except blocks.

Animated example of the for-loop

Sources:
Animating Python #3- For Loop: https://youtu.be/EfJVnAHiR4s

Take a tiny quiz on Python programming and feel good!

This quiz was graded!

Some rules to observe:

- 10 minutes restriction
- Unlimited responses
- You can discuss with your colleagues in-class
- Watch real-time results and calibrate

Skeleton of the updated Python program written for Raspberry Pi

```
import library1 as name1
import RPi.GPIO as GPIO
import time
```

Import libraries that are relevant for interaction with the Raspberry Pi hardware such as GPIO pins, camera ports etc.

```
GPIO.setmode(GPIO.BOARD)

GPIO.setup(12, GPIO.OUT)
```

Set up GPIO pins as data outputs or inputs for sensors and actuators

```
def function_name(arguments):
            statement
            statement
            . . .
            return value
```

Create user-defined functions to modularize your code and make it easy to work.

Create user-defined functions to release the GPIO pins

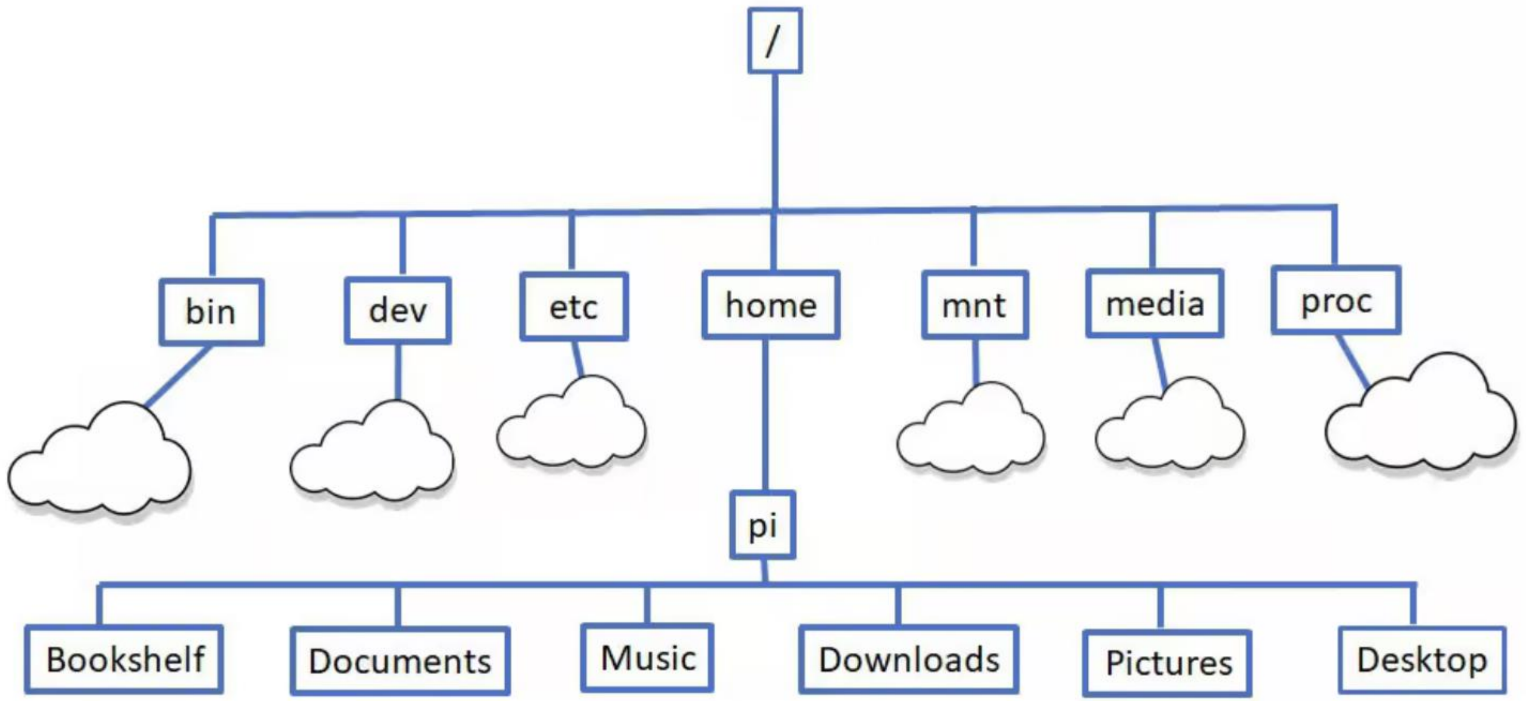
```
if __name__ == "__main__":
            try:
                        function_name1()
            except KeyboardInterrupt:
                        function_name2()
```

Create entry point into the program and pull all functions in

Create a keyboard-interrupt exception clause

The Linux Terminal

Source:
https://www.circuitbasics.com/how-to-navigate-the-raspberry-pi-file-system/



Explore files and folders

Command	Description
pwd	Show the **current** directory you are in
ls	List files in the current directory; add -l or -a for more detail or to see hidden files
cd /path/goes/here	Change directory (for example cd /home/pi/Documents)
mkdir myfolder	Create a new directory named myfolder
cp file1 file2	Copy a file; cp -r dir1 dir2 copies a whole directory
mv old new	Move or rename files/directories
rm file	Delete a file; use rm -r folder carefully to remove a directory

Helpful navigation and history tips

Command or Keyboard Entry	Description
clear	Clear the terminal screen.
history	Show a numbered list of past commands; run one again with !number.
Up/Down arrow keys	Scroll through previously used commands to run or edit them.

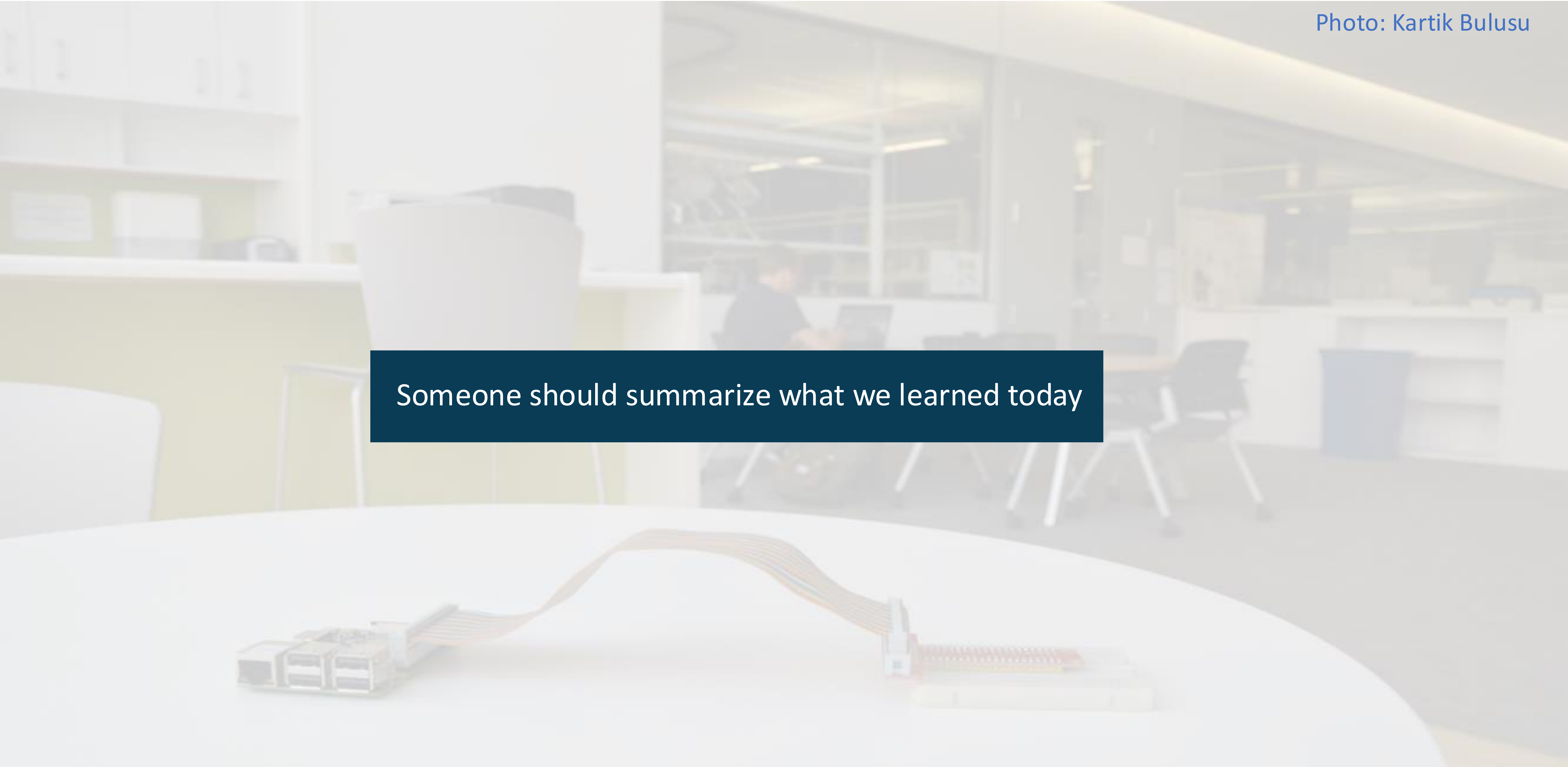
View and edit files

Command	Description
nano file.txt	Open a simple text editor in the terminal to create or edit a file
less file.txt	View a long file one page at a time (use q to quit)
cat file.txt	Print the contents of a text file to the screen

System information and status

Command	Description
hostname -I	Show the Pi’s IP address
uname -a	Show Linux kernel and system details
df -h	Show disk space usage in human-readable form
free -h	Show memory (RAM) usage
uptime	See how long the Pi has been running
top or htop	Monitor running processes and CPU usage (press q to exit top).

Photo: Kartik Bulusu



Someone should summarize what we learned today